SALTDRAG · RISK REGISTER AND HAZARD ANALYSIS

SaltDrag risk register and hazard analysis

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SDG-RSK-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdrag		
Source	docs/09-risk.md @ c7f0938		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	P5 mitigation as decided by Amish (D25), 2026-10-25
0.1	2026-10-24	Amish Chadha	First risk register, preliminary hazard analysis and first-pass design FMEA (kit 1.8 upgrade, stage P3)

SaltDrag risk register and hazard analysis

The top project risks are that the forest department treats a hand capstan as barred machinery, and that the rake drags harder on a real crystal bed than assumed; the top hazards are a loaded rope whipping back, heat stress in the harvest and hands caught in the cranks and gears. The highest design FMEA items are the rope (wear over the 100 mm roller and at the splices), the ground anchors in unknown soil and the stainless fittings in brine. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be found before harm) to 5 (unlikely to be found).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	The forest department does not allow the capstans in the sanctuary	Access conditions bar heavy machinery and new technology (SDG-REG-001 G2)	No trials in the Little Rann	3	5	Co-design partner	Written request before any build; fall back to coastal Kutch pans outside the sanctuary
P2	The rake drags much harder than 0.9 kN/m on a real bed	Layer loosening term assumed (SDG-STM-001 S1)	Crank force above 120 N, or a narrower rake and a lower rate	3	3	Design lead	Measure with a short trial rake and a spring scale before the first build is loaded
P3	SaltDrag gathers less per worker than the dantala	Baseline unknown; 133 kg per worker-hour estimated	PR-005 not met; the tool costs families time	3	4	Co-design partner	Time the dantala on the same pans first; compare in the first trial
P4	No co-design partner takes it up	Nothing agreed with SEWA or others	No trials, no users' voice in the design	3	4	Project lead	Approach SEWA and one other group with the costed set and the safety case
P5	One set per pan is not enough for a family	One set gathers about a third of a pan's yearly output in the harvest (SDG-CAL-001 [D12])	Families need two sets or a longer working season; cost per family doubles	3	3	Project lead	One set per pan kept as the design case (SDG-DEC-001 D25, decided by Amish 2026-10-25); ask the partner how long the gathering season really is and what share a family would gather
P6	Parts cost rises above the estimate	Indicative prices, first prototype	Less margin to the USD 4,000 target	2	2	Design lead	USD 1,103 of margin; prices confirmed on purchase (TRL 4)
P7	Soil behind the bunds too soft for the anchors	Saline silty soil; strength assumed 50 kPa (S12)	Capstan cannot be anchored	3	3	Design lead	Site pull test; deadweight or longer anchors as fallback

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Loaded rope whips back	Rope breaks, slips off the drum or loses its anchor	Severe injury to anyone in line with the rope	5	2	Low-stretch polyester at a working-load factor of 7.7; certified rope; damper bag on every loaded rope; 2 m exclusion zone along the rope line (SDG-DDR-001, D10)	Medium
H2	Hands caught in the gear mesh or struck by a crank	Unguarded mesh; crank spinning back	Crushed or broken fingers, struck forearm	4	2	Gear guard over the mesh; pawl always engaged while winding; cranks removed before paying out, which is done only on the brake (SDG-DDR-001, D9)	Low
H3	Capstan lurches toward the pan	Anchor pulls out under load	Struck or trapped crank workers	4	2	Two anchors per capstan proof-pulled on site to 3.42 kN (1.5 times a 70 % share); rated 10 mm chains at rope height (tipping margin 1.56, 1.46 at 6 kN)	Low
H4	Back strain or a dropped part when carrying	Parts up to 23.8 kg (ballast bag) and 23.5 kg (drum)	Back injury; crushed foot	3	3	Every part 25 kg or less; lift in pairs; capstan taken apart for moves	Medium
H5	Fall from the bund crest into the pan or onto the roller	Working near the 0.4 m crest	Bruising, sprain; contact with brine	2	2	Nobody needs to stand on the crest during a pass; the roller saddle is placed before the rope is loaded	Low
H6	Heat stress	Cranking at 64 to 71 W in March and April heat (SDG-STM-001 S16)	Heat exhaustion or heat stroke	5	3	Work in the early morning and evening; shade, water and rest at every strip change; crew rotation	Medium
H7	Welding and galvanising fumes, FRP dust in the workshop	Welding galvanised steel; drilling FRP	Metal fume fever; skin and lung irritation	3	2	Weld bare steel and galvanise afterwards; ventilation; dust mask and gloves for FRP (SDG-BLD-001)	Low
H8	Brine and salt on skin and eyes	Rigging the rake at the pan edge	Skin cracking, eye irritation	2	3	Rigging from the bund with the rope; gloves and eye protection; fresh water at hand	Low
H9	Brake band gives way while paying out	Worn or oily lining	Rake runs back; crank-free capstan spins	3	2	Paying out only with cranks removed; lining inspected daily; spare lining in the kit	Low

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Pull rope (BOM 18)	Carries up to 3 kN	Breaks at a splice or worn section	Snap-back (H1)	Wear over the 100 mm roller (D/d 8.3, at the low end; S27); UV; splice made badly	5	2	3	30	Certified rope with professional splices; daily inspection; replace at the first sign of glazing; check the maker's minimum D/d
F2	Helical anchors (BOM 11)	Hold the capstan	Pulls out	Capstan lurches (H3)	Soil softer than assumed (S12)	4	3	2	24	Proof pull on site before first use; reject and re-site any anchor that moves more than 25 mm
F3	Rake fittings, 316 stainless (BOM 15, 16, 19)	Hold blade, skids and bridles	Crevice corrosion under bolt heads	Loose blade or a bridle eye fails	Warm near-saturated brine, crevices (S19)	3	3	3	27	Daily rinse; inspect weekly; duplex 2205 bolts if pitting appears
F4	Pawl and ratchet (BOM 7)	Holds the drum when the cranks are let go	Pawl skips or its spring breaks	Drum runs back, crank spins (H2)	Salt in the teeth; spring corrosion	4	2	2	16	Spare pawl and spring in the kit; daily rinse; check engagement at every set-up
F5	Band brake (BOM 8)	Pays out under control	Slips	Rake runs back	Wet or salted lining	3	2	2	12	Cranks removed while paying out; lining kept dry and replaced when glazed
F6	Flange bearings (BOM 6)	Carry the shafts	Seize	Hard cranking	Salt dust in the seals	2	3	2	12	Stainless inserts, sealed; rinse and grease weekly
F7	Drum barrel (BOM 3)	Stores the rope	Crushes or ovalises	Rope binds, drum jams	Rope pressure under load (S7)	4	1	3	12	168.3 x 6.3 mm YSt 310 barrel, margin 2.17 at 6 kN (SDG-DEC-001 D18); proof load with CalRig at TRL 4
F8	Skids (BOM 15)	Set the blade 10 mm above the floor	Wear through	Blade digs into the floor (UR-004)	Abrasion on the crystal bed	3	3	2	18	Slotted hangers re-set the height; skids replaced from plastic stock
F9	Rake beam (BOM 13)	Carries the blade	Cracks at a bolt hole	Rake bends	FRP crushed by bolts; impact	3	2	3	18	Crush sleeves at every bolt; margin 7.1 at 6 kN
F10						3	2	2	12	

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
	Gear pair (BOM 4, 5)	Multiply the crank torque 6 to 1	Tooth wear or breakage	Hard cranking or loss of drive	Grit in the mesh					Guard over the mesh; grease weekly; Lewis margin 2.5 at 6 kN
F11	Anchor chain and turnbuckle (BOM 11b)	Tie the capstan to the anchors	Turnbuckle unscrews	Capstan creeps	Vibration while cranking	3	2	2	12	Lock nuts on the turnbuckles; check at every strip change